



BYD Microelectronics Co., Ltd.

BF30A2 芯片应用和设计指导

1/15" QVGA CMOS Image Sensor

BF30A2

芯片应用和设计指导



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1 外围电路

1.1 主板 IO 电源为 2.8V 时的原理图

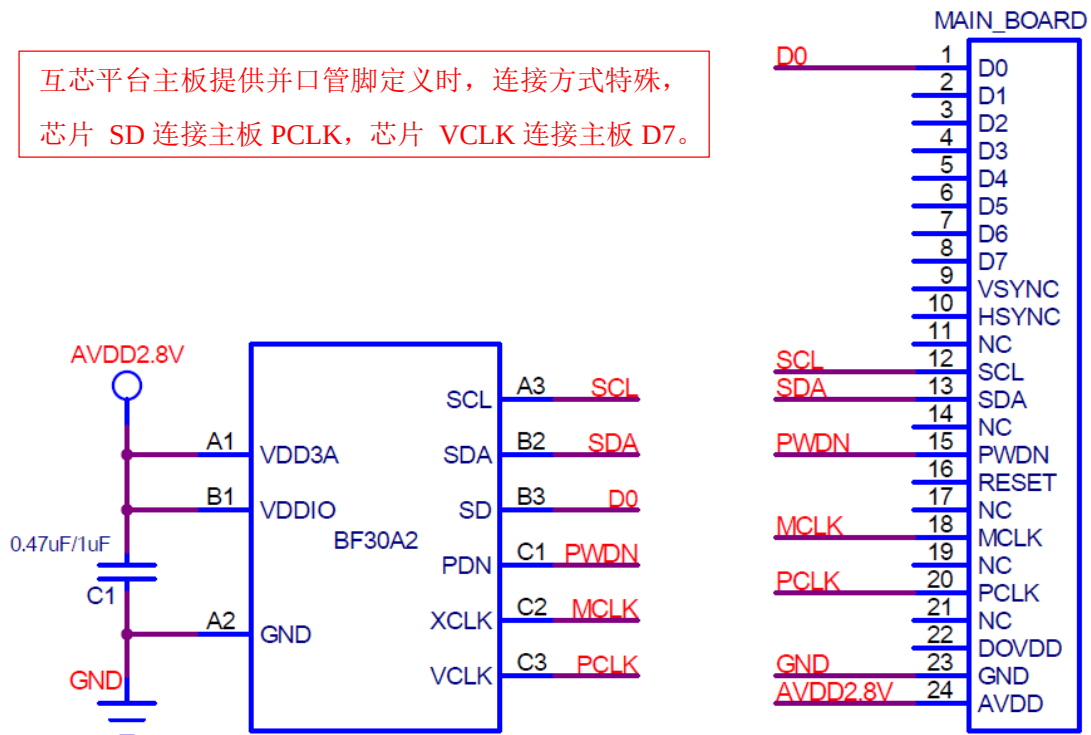


图 1-1 主板 IO 电源为 2.8V 时的原理图

1.2 主板 IO 电源为 1.8V 时的原理图

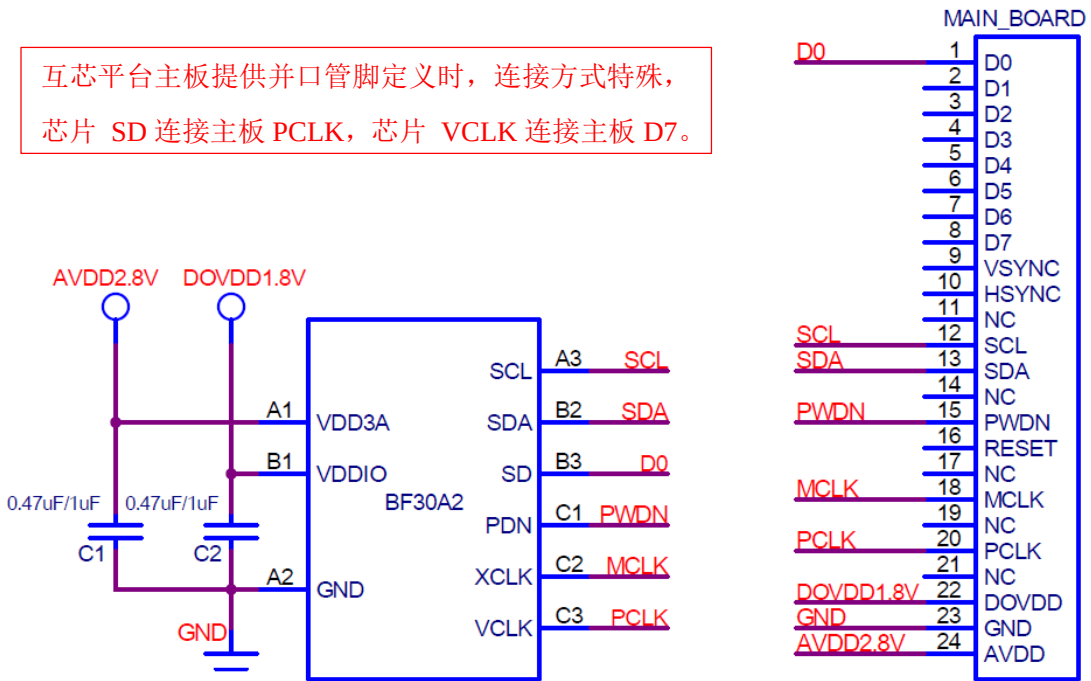


图 1-2 主板 IO 电源为 1.8V 时的原理图

1.3 外围电路设计说明

- BF30A2 芯片分两路电源供电：
VDDIO, 1.7 ~ 3.1V
VDD3A, 2.7 ~ 3.1V
- 主板 IO 电源 DOVDD 为 1.8V 时，芯片 VDDIO 必须接主板 DOVDD，不可与 VDD3A 共用电源；
- 建议电源和地线宽度 0.2mm 以上，空间允许时尽量加宽到 0.4mm，电源线尽量少打过孔，需先经过滤波电容再进入 sensor 的电源管脚，电容应尽量靠近对应电源管脚摆放；
- C1, C2 容值为 0.47 或 1uF，省略会影响图像质量；
- AVDD、MCLK、PCLK、D0 要尽量避免相邻并行走线，空间允许时建议分别包地。

2 PCB 焊盘设计示意图

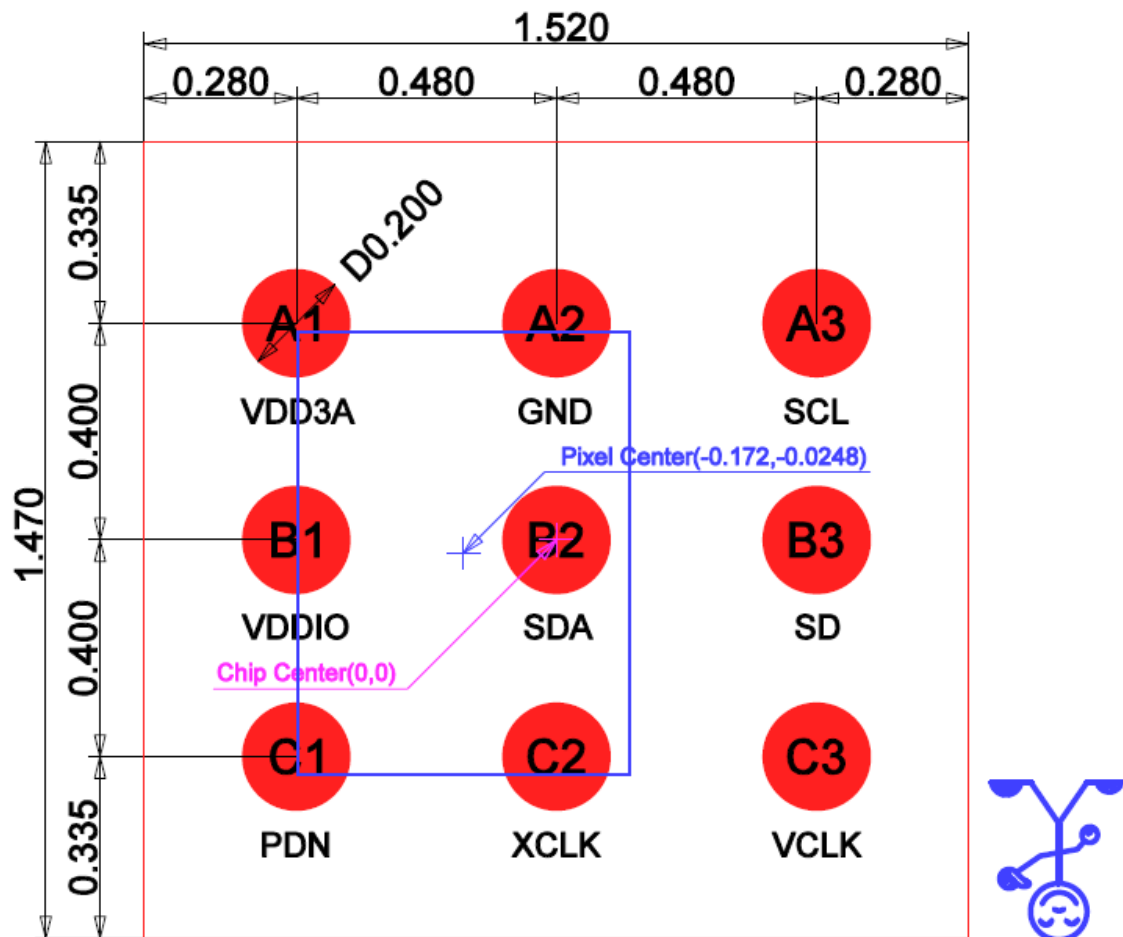


图 2-1 PCB 焊盘设计示意图 (单位: mm)

3 管脚说明

表 3-1 管脚说明

Pin Number	Name	Pin Type	Function/Description
A1	VDD3A	Supply	Analog power supply: 2.7V~3.1V
A2	GND	Ground	Ground
A3	SCL	Input	SCCB serial interface clock input
B1	VDDIO	Supply	I/O power supply: 1.7V~3.1V
B2	SDA	I/O	SCCB serial interface data inout
B3	SD	Output	Image data output port
C1	PDN	Input	Power Down Mode Selection. 0: Normal mode; 1: Power down mode.
C2	XCLK	Input	System input clock
C3	VCLK	Output	Pixel clock output

Notes:

1. PDN : pull-down to GND by inner resistor.

4 封装图

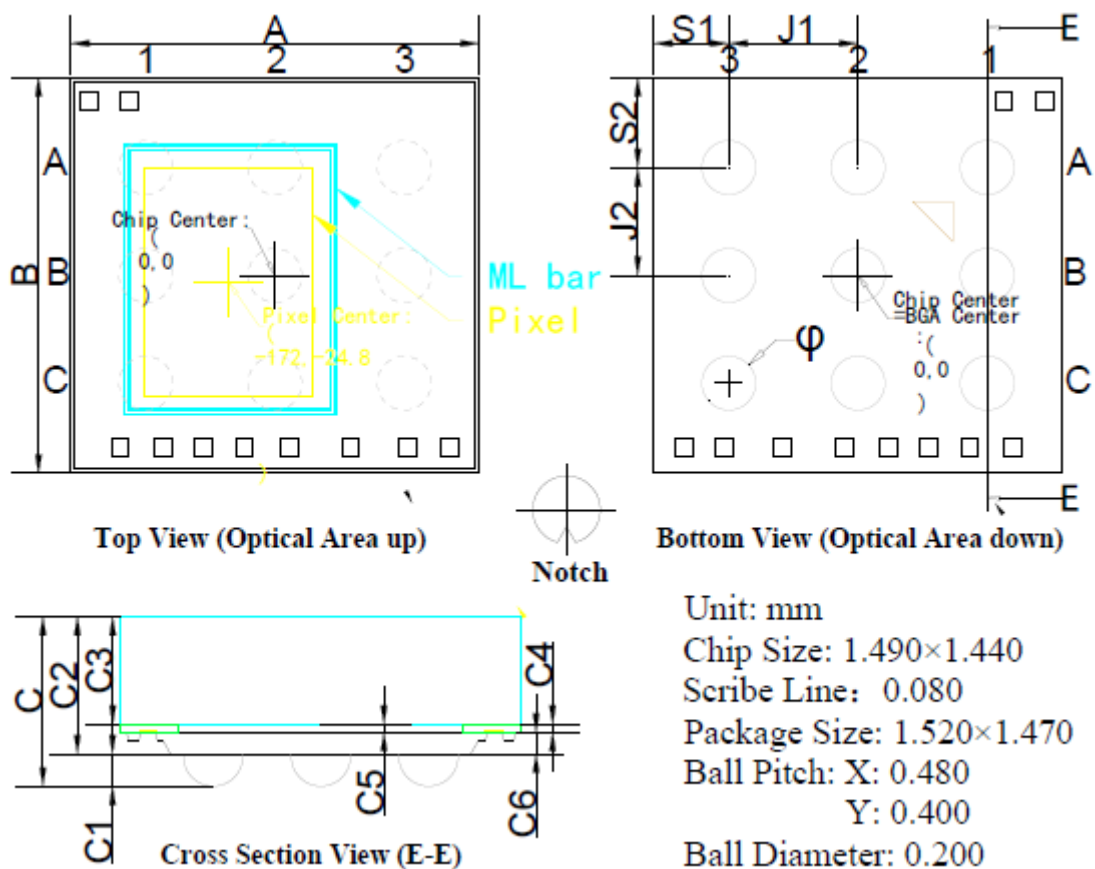


图 4-1 封装规格图

5 封装尺寸

表 5-1 封装尺寸

Description	Symbol	Nominal	Min	Max
		Millimeters		
Package Body Dimension X	A	1.5200	1.4950	1.5450
Package Body Dimension Y	B	1.4700	1.4450	1.4950
Package Height	C	0.6500	0.5950	0.7050
Ball Height	C1	0.1000	0.0700	0.1300
Package Body Thickness	C2	0.5500	0.5200	0.5800
Glass Thickness	C3	0.4000	0.3900	0.4100
Cavity wall height	C4	0.0300	0.0260	0.0340
Cavity Wall + Epoxy Thickness	C5	0.0325	0.0275	0.0375
Si Thickness	C6	0.0800	0.0750	0.0850
Ball Diameter	D	0.2000	0.1700	0.2300
Total Ball Count	N	9		
Pins pitch X axis	J1	0.4800		
Pins pitch Y axis	J2	0.4000		
Edge to Ball Center Distance along X	S1	0.2800	0.2500	0.3100
Edge to Ball Center Distance along Y	S2	0.3350	0.3050	0.3650

6 回流焊条件

BF30A2 为无铅封装器件，其锡球成分为：Sn:96.5%；Ag: 3%；Cu: 0.5%。回流焊工序推荐选用无铅制程。

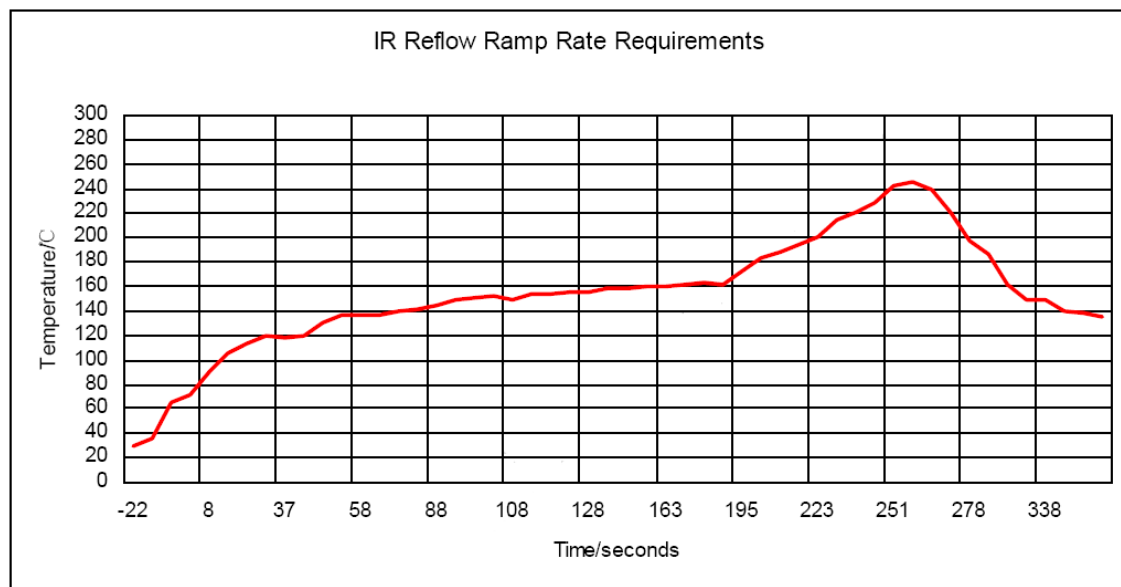


图 6-1 回流焊温度曲线

表 6-1 回流焊温度分布



温度	时间
斜升速率 (30°C - 217°C)	小于 3°C /秒
> 100°C	控制在 330 - 600 秒之间
> 150°C	不少于 210 秒
> 217°C	不少于 30 秒 (30 - 120 秒)
峰值温度	245°C
斜降速率 (245°C - 50°C)	小于 6°C/秒
30°C - 245°C	最大值 390 秒

7 元件车间寿命

BF30A2 属于 2a 级湿敏元件，在不同温湿度下的车间寿命请参考下表（根据 J-STD-033C，BYD MSL=2a）：

表 7-1 元件车间寿命与温湿度对应表

	5%	10%	20%	30%	40%	50%	60%	70%	80%	90%
35°C	∞	∞	∞	∞	∞	∞	17	1	0.5	0.5
30°C	∞	∞	∞	∞	∞	∞	28	1	1	1
25°C	∞	∞	∞	∞	∞	∞	∞	2	1	1
20°C	∞	∞	∞	∞	∞	∞	∞	2	2	1

备注：1. 上表中湿度为相对湿度，车间寿命单位为天；

2. 当芯片暴露时间超过车间寿命时，SMT 上线前必须对芯片进行烘烤。

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